

Lesson 9

Euclidean Algorithm, Linear Diophantine Equations

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9.1 The Euclidean Algorithm

As you see in the examples and as your experiences, greatest common divisor of two integers can be found by listing all their positive divisors and choosing the largest one common to each. But this method is cumbersome for finding greatest common divisor of large numbers. Using the repeated application of Division Algorithm an efficient method called Euclidean algorithm can be developed to find greatest common divisor.

Before introducing the Euclidean algorithm, let's establish a lemma

9.1.1 Lemma

If $a = qb + r$, then $\gcd(a, b) = \gcd(b, r)$

Proof:

Let $d = \gcd(a, b)$.

Then since $d|a$ and $d|b$, $d|(a - qb)$

Thus d is a common divisor of b and r .

Now let c be an arbitrary positive common divisor of b and r .

Therefore $c|(qb + r)$.

Thus $c|a$

Hence c is a common divisor of a and b .

Therefore since $d = \gcd(a, b)$, $c \leq d$.

It now follows that from the definition of $\gcd(b, r)$, that $d = \gcd(a, b) = \gcd(b, r)$. ■

Now with the help of above lemma we can establish the Euclidean Algorithm.

9.1.2 Theorem -Euclidean Algorithm

Given integers a and $b > 0$, there exist a series of equations

$$\begin{aligned}a &= bq_1 + r_1, & 0 < r_1 < b \\b &= bq_2 + r_2, & 0 < r_2 < r_1 \\r_1 &= r_2q_3 + r_3, & 0 < r_3 < r_2 \\&\vdots \\r_{n-2} &= r_{n-1}q_n + r_n, & 0 < r_n < r_{n-1} \\r_{n-1} &= r_nq_{n+1} + 0\end{aligned}$$

such that $\gcd(a, b) = r_n$.

Proof:

Using the result of above lemma we can simply write the equations

$$\gcd(a, b) = \gcd(b, r_1) = \cdots = \gcd(r_{n-1}, r_n) = \gcd(r_n, 0) = r_n. \blacksquare$$

9.1.3 Example

Find $\gcd(42823, 6409)$

Answer:

$$42823 = 6 \cdot 6409 + 4369$$

$$6409 = 1 \cdot 4369 + 2040$$

$$4369 = 2 \cdot 2040 + 289$$

$$2040 = 7 \cdot 289 + 17$$

$$289 = 17 \cdot 17.$$

Hence $\gcd(42823, 6409) = 17$.

Euclidean algorithm is not only gives a fast method to find the greatest common divisor, but also give a method to find exact integers x and y such that $\gcd(a, b) = ax + by$. For an example find the integers x and y such that $\gcd(42823, 6409) = 17 = 42823x + 6409y$ in the above example.

We can start with the one line before the last line of the Euclidean algorithm.

$$\text{Hence } 17 = 2040 - 7.289$$

$$289 = 4369 - 2.2040$$

$$2040 = 6409 - 4369$$

$$4369 = 42823 - 6.6409$$

From these equations,

$$\begin{aligned} 17 &= 2040 - 7.289 \\ &= 2040 - 7(4369 - 2.2040) \\ &= -7.4369 + 15.2040 \\ &= -7(42823 - 6.6409) + 15.(6409 - 4369) \\ &= -7.42823 + 57.6409 - 15.4369 \\ &= -7.42823 + 57.6409 - 15(42823 - 6.6409) \\ &= -22.42823 + 147.6409 \end{aligned}$$

$$\text{Hence } x = -22, y = 147.$$

9.2 Linear Diophantine Equation $ax + by = c$

If we looking for real solutions for the equation $ax + by = c$ where a, b , and c are real numbers, then what we can say about the set of solutions? We can write the set of solutions as

$S = \{(x, y) | ax + by = c, x, y \in \mathbb{R}\}$. Notice that if $c \neq 0$ and $a = b = 0$ then the set S is empty. In all other cases the set S is infinite.

If a, b, c, x and y are integers then the equation $ax + by = c$ is called Linear Diophantine Equation. Finding the integer solutions of the Linear Diophantine Equation is different from finding real solutions. Consider the following two Linear Diophantine Equations.

$$(i) \quad 2x + 5y = 64$$

$$(ii) \quad 2x + 10y = 63$$

One can easily see that the first equation has a solution $x = 7, y = 10$. Also $x = 2, y = 12$ is another solution. If you give more attention to the integer solutions, you can realize that there are infinitely many solutions to this equation. In the second equation, while the right hand side is an odd number, the left hand side is always an even number. This argument shows that the second equation does not have any integer solutions. The following theorem gives condition for solvability. Also it gives all the solutions if one solution is known.

9.2.2 Theorem

The linear Diophantine equation $ax + by = c$ has a solution if and only if $d|c$, where $d = \gcd(a, b)$. If (x_0, y_0) is any particular solution of this equation, then all other solutions are given by $x = x_0 + \left(\frac{b}{d}\right)t, y = y_0 - \left(\frac{a}{d}\right)t$, where t is an arbitrary integer.

Proof:

Suppose $ax + by = c$ has a solution.

Let $d = \gcd(a, b)$.

Therefore we know that there are integers r and s such that $a = dr$ and $b = ds$.

Let (x_1, y_1) be a solution. Therefore $c = ax_1 + by_1 = drx_1 + dsy_1 = d(rx_1 + sy_1)$.

Which says that $d|c$.

Conversely suppose that $d|c$. Hence there is an integer t such that $c = dt$. By Theorem 9.2.2, there are integers x_0 and y_0 such that $d = ax_0 + by_0$.

Hence $c = dt = (ax_0 + by_0)t = a(tx_0) + b(ty_0)$.

Therefore the Diophantine equation $ax + by = c$ has a particular solution.

Now to prove the second assertion of the theorem, suppose (x_0, y_0) is a known solution of $ax + by = c$.

If (x', y') is any other solution then $ax_0 + by_0 = c = ax' + by'$.

Hence $a(x' - x_0) = -b(y_0 - y')$

By dividing both sides by $d = \gcd(a, b)$, $\frac{a}{d}(x' - x_0) = -\frac{b}{d}(y_0 - y')$ {Notice that $\frac{a}{d}$ and $\frac{b}{d}$ are integers and $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$ }

Therefore $\frac{a}{d} | (y_0 - y')$ and $\frac{b}{d} | (x' - x_0)$.

Hence there is an integer t such that $x' = x_0 + \left(\frac{b}{d}\right)t$, $y' = y_0 - \left(\frac{a}{d}\right)t$.

Also if (x', y') is in this form then it is a solution because,

$$ax' + by' = ax_0 + \frac{abt}{d} + by_0 - \frac{abt}{d} = c. \blacksquare$$

9.2.3 Example

Find the solutions of each of the following Diophantine equations if those exist.

(i) $3x + 2y = 5$

(ii) $21x + 111y = 7$

Answers : Since $\gcd(3, 2)=1$ and $1|5$, the equation $3x + 2y = 5$ is solvable. Also observe that $x_0 = 1$ and $y_0 = 1$ is a solution. Hence all possible solutions are of the form $x = 1 + 2t$, $y = 1 - 3t$, $t \in \mathbb{Z}$.

In the second question, the equation $21x + 111y = 7$ has no solution, since $\gcd(21, 111)=3$ and $3 \nmid 7$.

Review Questions

(1) Use Euclidean Algorithm to find $\gcd(987, 144)$. Find $x, y \in \mathbb{Z}$ such that $\gcd(987, 144) = 987x + 144y$.

(2) A house wife spends 1770 Rupees for purchasing mangoes and apples. She pays 31 Rupees for each mango and 21 Rupees for each apple. If she buys more mangoes than apples, how many mangoes and apples she buys?

(3) Use Euclidean Algorithm to find $x, y \in \mathbb{Z}$ such that $34x + 21y = 1$.

Grade 4+

අනිවාර්ය අන්දම 21 - Business in Dolars



In a certain country, as their currency, people using only two kinds of dollar bills, \$34 and \$55. From a toy shop a person need to buy a toy for \$4.

By assuming both the buyer & seller have plenty of dollar bills of both kinds, suggest a method to exchange money, after buying the toy.

Grade 4+

අන්තර් අන්තර් 21 - බොල් ගණු දෙනු



එකෝමන් එක රටක ව්‍යවහාර මුදල් විභින්න භාවිතා කරන්නේ \$55 සහ \$34 වටිනාකම් ඇති මුදල් නෝට්ටු වර්ග දෙක පමණයි. මේ රටේ සෙල්ලම් බඩු කඩයකින් මිනිසෙකුට \$4 ක් වන සෙල්ලම් බඩුවක් මිලදී ගැනීමට අවශ්‍ය වුනා.

මිලදී ගන්නාට සහ විකුණන්නාට ඇති පද්ධි වර්ග දෙ-වර්ගයේම බොල් නෝට්ටු ඇතැයි උපකල්පනය කරමින් සෙල්ලම් බඩුව මිලදී ගත් පසු මුදල් හුවමාරු කර ගැනීමට ක්‍රමයක් යෝජනා කරන්න.